

# Functional Requirements Specification (FRS)

**System:** Jima — CARAVAN Lounge Restaurant Management System  
**Document:** FRS v1.2 · **Updated:** August 20, 2026  
**Related documents:** SRS (02), SDD (03), User Manual (04), Handoff (05), deployment guide (06), and operational runbooks (07–12)

## 1. Purpose

This document defines the functional requirements of the Jima Restaurant Management System — a full-stack web application that manages the daily operations of CARAVAN Lounge: ordering, kitchen workflow, payments, inventory, staff, branches, and reporting.

**Verified implementation baseline:** the registered Jima web artifact runs the real Next.js 15 / React 19 frontend in `artifacts/nextjs-app`. Browser requests use `/backend/api/*`, which Next.js rewrites to the NestJS service's `/api/*`. NestJS and TypeORM own all PostgreSQL access. The separate `artifacts/api-server` is not the restaurant application API.

**Requirements versus evidence:** the items below define required behavior. A requirement must not be treated as production-verified unless it appears as passed evidence in document 12.

## 2. User Roles

| Role  | Description                                                     |
|-------|-----------------------------------------------------------------|
| Admin | Full system access; manages all data, users, and configuration. |

|             |                                                                                        |
|-------------|----------------------------------------------------------------------------------------|
| Owner       | Business owner; same operational visibility as manager, including analytics dashboard. |
| Manager     | Runs daily operations; manages menu, staff, tables, branches, inventory, and reports.  |
| Coordinator | Confirms incoming orders and assigns them to chefs; can cancel orders.                 |
| Waiter      | Takes orders at tables (POS), adds items to open orders, tracks order progress.        |
| Chef        | Works the kitchen board: accepts, prepares, and marks orders ready.                    |
| Cashier     | Collects payments, closes orders, views sales and items-purchased reports.             |
| Storekeeper | Manages stock levels, receives goods, handles item requests and adjustments.           |

## 3. Functional Requirements

### 3.1 Authentication & Authorization

| ID            | Requirement                                                                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FR-A01</b> | Users log in with e-mail and password; the system returns a JWT access token.                                                                                                                           |
| <b>FR-A02</b> | Every protected API call requires a valid <code>Authorization: Bearer</code> token. Login and health endpoints are intentional public exceptions; administrative seed operations must remain protected. |
| <b>FR-A03</b> | Menu entries and pages are restricted per role (role-based access control).                                                                                                                             |
| <b>FR-A04</b> | Passwords are stored hashed (bcrypt), never in plain text.                                                                                                                                              |

### 3.2 Orders & POS

| ID | Requirement |
|----|-------------|
|----|-------------|

|               |                                                                                                                                                                                                                                      |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FR-O01</b> | Waiters create orders through a POS screen: pick a table (or take-away), select menu items by category, set quantities and notes.                                                                                                    |
| <b>FR-O02</b> | Orders follow the lifecycle: pending → confirmed → preparing → ready → served → paid , with cancelled possible before "ready".                                                                                                       |
| <b>FR-O03</b> | The coordinator confirms a pending order and assigns a chef to it.                                                                                                                                                                   |
| <b>FR-O04</b> | Waiters may append items to any order that is not yet paid or cancelled.                                                                                                                                                             |
| <b>FR-O05</b> | The orders list shows table, chef, item count, time, total, payment method, and status; rows open a detail view.                                                                                                                     |
| <b>FR-O06</b> | Orders can be filtered by status, waiter, table, item, date, time, and payment method.                                                                                                                                               |
| <b>FR-O07</b> | The list and open detail view refresh automatically every 10 seconds.                                                                                                                                                                |
| <b>FR-O08</b> | On phone-sized screens, the POS keeps menu and checkout controls usable by stacking the menu above the cart, preserving horizontal category browsing, and providing separate vertical scrolling for menu items and checkout content. |

### 3.3 Kitchen

| ID            | Requirement                                                                                                                |
|---------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>FR-K01</b> | The kitchen board shows confirmed orders queued per chef, with tabs per chef.                                              |
| <b>FR-K02</b> | Chefs move orders through accept → preparing → ready.                                                                      |
| <b>FR-K03</b> | The Order Progress Board displays four live columns: Requested, Received & Confirmed, In Progress, Completed (ready only). |
| <b>FR-K04</b> | Orders waiting more than 20 minutes are visually flagged.                                                                  |

### 3.4 Payments

| ID            | Requirement |
|---------------|-------------|
| <b>FR-P01</b> |             |

|               |                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------|
|               | Cashiers record payments against served orders with method cash, card, or wallet, including change given. |
| <b>FR-P02</b> | Paying an order marks it <b>paid</b> and frees its table.                                                 |
| <b>FR-P03</b> | Daily reports summarize revenue, transaction counts, and per-method breakdowns.                           |

### 3.5 Menu Management

| ID            | Requirement                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------|
| <b>FR-M01</b> | Managers maintain menu categories and items (name, price, description, availability, preparation time). |
| <b>FR-M02</b> | Unavailable items are hidden from the POS.                                                              |

### 3.6 Inventory & Store

| ID            | Requirement                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>FR-I01</b> | Inventory items track unit, current stock, minimum stock, reorder point, unit cost, category, and expiry date.                          |
| <b>FR-I02</b> | Low-stock alerts flag items below minimum (critical when at or below half the minimum, or zero).                                        |
| <b>FR-I03</b> | Suppliers and purchase orders are managed with status tracking.                                                                         |
| <b>FR-I04</b> | Stock adjustments record additions, deductions, and waste with reasons and the acting user.                                             |
| <b>FR-I05</b> | Staff submit item requests; managers/storekeepers approve, issue, and track them, with low-stock / expiry badges shown on request rows. |

### 3.7 Staff, Branches & Tables

| ID            | Requirement                                                                 |
|---------------|-----------------------------------------------------------------------------|
| <b>FR-S01</b> | Managers create and manage staff accounts with roles and branch assignment. |

|               |                                                                                |
|---------------|--------------------------------------------------------------------------------|
| <b>FR-S02</b> | Restaurants and branches are managed hierarchically (restaurant → branches).   |
| <b>FR-S03</b> | Tables have number, capacity, section, and live status (available / occupied). |

### 3.8 Dashboards & Reports

| ID            | Requirement                                                                                                                                                                                                                                              |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FR-D01</b> | Admin/Owner/Manager see a combined dashboard: today's KPIs (sales, orders, average order value, in-progress, low stock, stock value), sales trend, top sellers, profitability, and kitchen performance — on a single page. KPI cards open detail popups. |
| <b>FR-D02</b> | Reports offer per-area export tabs (Order Board, Menu, Inventory, Item Requested, Staff, Branches) with filters and Excel / PDF export.                                                                                                                  |
| <b>FR-D03</b> | Cashiers see a reduced report view: Sales and Items Purchased only, filterable by type and date, exportable to Excel / PDF.                                                                                                                              |
| <b>FR-D04</b> | Role-specific landing: waiters, coordinators, and cashiers land on Orders/POS instead of the dashboard.                                                                                                                                                  |

### 3.9 Notifications

| ID            | Requirement                                                                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>FR-N01</b> | The system generates notifications targeted at roles or specific users (e.g. new order, order ready) with read tracking. |

## 4. Business Rules

- An order's total is the sum of its items (quantity × unit price) plus any service charge percentage.
- Only pending orders can be confirmed; only served orders can be paid.
- Cancelling is allowed only while an order is pending, confirmed, or preparing.

- Data is scoped by branch where applicable (orders, tables, staff, stock).

## 5. Verification Status

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Functional development is active and major workflows have been exercised manually. Complete automated regression evidence for every role, branch boundary, order transition, payment, inventory operation, notification, report, and export is not yet recorded. Production release remains blocked until the verification matrix in document 12 is complete.